DESIGN AND ANALYSIS
OF
ALGORITHMS LAB
[AI286]

A File submitted in partial fulfilment of the

Requirements for the award of the degree of

B. TECH

IN

COMPUTER SCIENCE AND ENGINEERING(AI)

Submitted by:

Deepak Pal

235/UAI/014

Submitted to:

Dr. Gaurav Kumar

(Assistant Professor)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING,

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY,

GAUTAM BUDDHA UNIVERSITY, GREATER NOIDA-201312 GAUTAM BUDH NAGAR, UTTAR PRADESH, INDIA 2024-2025

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1.Sum of N numbers:

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Name: Deepak pal\nRoll No: 235UAI014\n");
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        sum += i;
    }
    printf("Sum of first %d natural numbers is: %d\n", n, sum);
    return 0;
}
```

Output

```
Name: Deepak pal
Roll No: 235UAI014
Enter a positive integer: 10
Sum of first 10 natural numbers is: 55


=== Code Execution Successful ===
```

2. Bubble Sort:

```
#include <stdio.h>
void bubbleSort(int arr[], int n) {
    for (int i = 0; i < n - 1; i++)
        for (int j = 0; j < n - i - 1; j++)
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
}

int main() {
    int arr[] = {6, 5, 4, 3, 2, 1};
    int n = sizeof(arr) / sizeof(arr[0]);

    printf("Name: Deepak pal\nRoll No: 235UAI014\n");

    bubbleSort(arr, n);
    printf("Sorted array: ");
    for (int i = 0; i < n; i++) printf("%d ", arr[i]);
```

```
    return 0;
}
```

Output

```
Name: Deepak pal
Roll No: 235UAI014
Sorted array: 1 2 3 4 5 6

=== Code Execution Successful ===
```

3.. Selection Sort:

```
 #include <stdio.h>


void selectionSort(int arr[], int n) {
   for (int i = 0; i < n - 1; i++) {
      int minIndex = i;
      for (int j = i + 1; j < n; j++) {
         if (arr[j] < arr[minIndex]) {
            minIndex = j;
         }
      }
      // Swap the found minimum element with the first
element
      int temp = arr[minIndex];
      arr[minIndex] = arr[i];
      arr[i] = temp;
   }
}


void printArray(int arr[], int n) {
   for (int i = 0; i < n; i++)
```

```
        printf("%d ", arr[i]);
    printf("\n");
}

int main() {
printf(“Name: Deepak pal\nRoll no: 235UAI014\n”);
    int arr[] = {64, 25, 12, 22, 11};
    int n = sizeof(arr) / sizeof(arr[0]);

    selectionSort(arr, n);
    printf("Sorted array: \n");
    printArray(arr, n);
    return 0;
```

Output

```
Name: Deepak pal
Roll no: 235UAI014
Sorted array:
11 12 22 25 64


=== Code Execution Successful ===
```

4. Linear Search

```
#include <stdio.h>
int linearSearch(int arr[], int n, int target) {
    for (int i = 0; i < n; i++) {
        if (arr[i] == target) return i;
    }
    return -1;
}
int main() {
    printf("Name: Deepak Pal I\nRoll No: 235UAI014\n");
    int n, target;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d elements: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the element to search: ");
    scanf("%d", &target);
    int result = linearSearch(arr, n, target);
    if (result != -1)
        printf("Element found at index %d\n", result);
    else
        printf("Element not found\n");
```

```
    return 0;
}
```

Output

```
Name: Deepak Pal I
Roll No: 235UAI014
Enter the number of elements: 6
Enter 6 elements: 1
2
3
4
5
6
Enter the element to search: 3
Element found at index 2


=== Code Execution Successful ===
```

5. Binary search using recursion

```
#include <stdio.h>
int binarySearch(int arr[], int left, int right, int target) {
    if (left > right) return -1;
    int mid = left + (right - left) / 2;
    if (arr[mid] == target) return mid;
    else if (arr[mid] > target) return binarySearch(arr, left, mid - 1, target);
    else return binarySearch(arr, mid + 1, right, target);
}
int main() {
    printf("Name: Deepak Pal\nRoll No: 235UAI014\n");
    int n, target;
    printf("Enter the number of elements: ");
    scanf("%d", &n);
    int arr[n];
    printf("Enter %d sorted elements: ", n);
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }
    printf("Enter the element to search: ");
    scanf("%d", &target);
    int result = binarySearch(arr, 0, n - 1, target);
    if (result != -1)
        printf("Element found at index %d\n", result);
```

```
    else
        printf("Element not found\n");
    return 0;
}
```

Output

```
Name: Deepak Pal
Roll No: 235UAI014
Enter the number of elements: 5
Enter 5 sorted elements: 1
2
3
4
5
Enter the element to search: 2
Element found at index 1


=== Code Execution Successful ===
```

6. Print number in reverse order through recursion

```
#include <stdio.h>
void printReverse(int n) {
    if (n < 1) return;
    printf("%d ", n);
    printReverse(n - 1);
}
int main() {
    printf("Name: Deepak Pal\nRoll No: 235UAI014\n");
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);
    printf("Numbers in reverse order: ");
    printReverse(n);
    return 0;
```

Output

```
Name: Deepak Pal
Roll No: 235UAI014
Enter a number: 6
Numbers in reverse order: 6 5 4 3 2 1

=== Code Execution Successful ===
```

}

7. Insertion sort

```
#include <stdio.h>

void insertionSort(int arr[], int n) {
    for (int i = 1; i < n; i++) {
        int key = arr[i], j = i - 1;
        while (j >= 0 && arr[j] > key) arr[j + 1] = arr[j], j--;
        arr[j + 1] = key;
        for (int k = 0; k < n; k++) printf("%d ", arr[k]);
        printf("\n");
    }
}

int main() {
    printf("Name: Deepak Pal\nRoll No: 235UAI014\n");
    int arr[] = {12, 11, 13, 5, 6}, n = 5;
    insertionSort(arr, n);
    return 0;
```

Output

```
Name: Deepak Pal
Roll No: 235UAI014
11 12 13 5 6
11 12 13 5 6
5 11 12 13 6
5 6 11 12 13


=== Code Execution Successful ===
```

}

8. Quick sort

```
#include <stdio.h>

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}

int partition(int arr[], int low, int high) {
    int pivot = arr[high]; // Choosing last element as the pivot
    int i = (low - 1);
    for (int j = low; j < high; j++) {
        if (arr[j] < pivot) {
            i++;
            swap(&arr[i], &arr[j]);
        }
    }
    swap(&arr[i + 1], &arr[high]);
    return (i + 1);
}

void quickSort(int arr[], int low, int high) {
    if (low < high) {
        int pi = partition(arr, low, high);
        quickSort(arr, low, pi - 1); // Sort left partition
        quickSort(arr, pi + 1, high); // Sort right partition
    }
```

```c
}

int main() {
    printf("Name: Deepak Pal \nRoll No: 235UAI014n");
    int arr[] = {10, 7, 8, 9, 1, 5};
    int n = sizeof(arr) / sizeof(arr[0]);
    printf("Original array: ");
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    quickSort(arr, 0, n - 1);
    printf("\nSorted array: ");
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```

Output

```
Name: Deepak Pal
Roll No: 235UAI014
Original array: 10 7 8 9 1 5
Sorted array: 1 5 7 8 9 10

=== Code Execution Successful ===
```

9. Factorial with recursion

```
#include <stdio.h>
int factorial(int n) {
   if (n == 0 || n == 1)
      return 1;
   return n * factorial(n - 1);
}
int main() {
   int n;
   printf("Name Deepak Pal \nRoll No: 235UAI014\n");
   printf("Enter a positive integer: ");
   scanf("%d", &n);
   printf("Factorial of %d is: %d\n", n, factorial(n));
   return 0;
}
```

Output

```
ame Deepak Pal
oll No: 235UAI014
nter a positive integer: 5
actorial of 5 is: 120


== Code Execution Successful ===
```

10. Fibonacci series by recursion

```c
#include <stdio.h>

int fibonacci(int n) {

    if (n <= 1)

        return n;

    return fibonacci(n - 1) + fibonacci(n - 2);

}

int main() {

    printf("Name: Deepak Pal\nRoll No: 235UAI014\n");

    int n;

    printf("Enter the number of terms: ");

    scanf("%d", &n);

    printf("Fibonacci Series: ");

    for (int i = 0; i < n; i++) {

        printf("%d ", fibonacci(i));

    }

    return 0;

}
```

Output

```
Name: Deepak Pal
Roll No: 235UAI014
Enter the number of terms: 4
Fibonacci Series: 0 1 1 2

=== Code Execution Successful ===
```